

DBMS:

Q1) Create a database named CollegeDB_Your_Roll_Number.

```
mysql> CREATE DATABASE CollegeDB_50N;
Query OK, 1 row affected (0.003 sec)

mysql> use DATABASE CollegeDB_50N;
ERROR 1049 (42000): Unknown database 'database'
mysql> use CollegeDB_50N;
Database changed
```

Q2) Create a table named Students in CollegeDB_Your_Roll_Number Database with the following columns (no constraints applied, you have to identify appropriate datatypes for columns and apply):

- StudentID
- Name
- Dept
- Age
- City

```
mysql> create table Students(StudentID int,Name varchar(10),Dept varchar(10),Age int,City varchar(10));
Query OK, 0 rows affected (0.008 sec)

mysql> show table;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
mysql> show table Students;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'Students' at line 1
mysql> desc table Students;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | Students | NULL | ALL | NULL | NULL | NULL | NULL | 1 | 100.00 | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set, 1 warning (0.004 sec)
```

Task 2: Modifying Table Structure

Q3) Add two new columns to the Students table:

- Email
- Phone

```
mysql> alter table Students;
Query OK, 0 rows affected (0.001 sec)

mysql> ALTER TABLE Students
    -> ADD Email VARCHAR(50),
    -> ADD Phone VARCHAR(15);
Query OK, 0 rows affected (0.012 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Q4) Modify the datatype of the column Phone to BIGINT

```
mysql> alter table Students
[    -> modify Phone bigint;
Query OK, 0 rows affected (0.014 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q5) Rename column Phone to ContactNumber.

```
mysql> alter table Students
    -> change Phone ContactNumber bigint;
Query OK, 0 rows affected (0.006 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q6) Remove the column Email from the Students table.

```
[mysql> alter table Students
[    -> drop column Email;
Query OK, 0 rows affected (0.013 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q7) Change the table name from Students table to Student_info table.

```
[mysql> rename table Students to Student_info;
[Query OK, 0 rows affected (0.007 sec)
```

Task 3: Inserting and Observing Data

Q8) Insert at least 5 records into the Student_info table.

- Try inserting duplicate values for StudentID.
- Try inserting NULL values in Age or Dept.

```
mysql> insert into Student_info values(1,'Arjun','CSE',20,'Hyderabad',38947589);
Query OK, 1 row affected (0.003 sec)
[
mysql> select * from Students
    -> ;
ERROR 1146 (42S02): Table 'collegedb_123.students' doesn't exist
mysql> select * from Sudent_info;
ERROR 1146 (42S02): Table 'collegedb_123.sudent_info' doesn't exist
mysql> select * from Student_info;
+-----+-----+-----+-----+-----+-----+
| StudentID | Name  | Dept | Age | City       | ContactNumber |
+-----+-----+-----+-----+-----+-----+
|          1 | Arjun | CSE  | 20  | Hyderabad | 38947589      |
+-----+-----+-----+-----+-----+-----+
[1 row in set (0.000 sec)
[
mysql> insert into Student_info values(2, 'Meena', 'ECE', 21, 'Delhi', 9123456780);
```

```
mysql> insert into Student_info values(2, 'Meena', 'ECE', 21, 'Delhi', 9123456780);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(2, 'Meena', 'ECE', 21, 'Delhi', 9123456780);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(3, 'Ravi', NULL, 19, 'Chennai', 9988776655);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(4, 'Sita', 'IT', NULL, 'Mumbai', 9876501234);
Query OK, 1 row affected (0.002 sec)

mysql> insert into Student_info values(5, 'Kiran', 'CSE', 20, 'Hyderabad', 9876543211);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(6, 'Deepa', 'EEE', 22, 'Delhi', 9345678901);
Query OK, 1 row affected (0.001 sec)
```

Q9) Insert at least 5 records at once into the Student_info table using a single INSERT statement.

```
mysql> insert into Student_info values(4, 'Sita', 'IT', NULL, 'Mumbai', 9876501234);
Query OK, 1 row affected (0.002 sec)

mysql> insert into Student_info values(5, 'Kiran', 'CSE', 20, 'Hyderabad', 9876543211);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(6, 'Deepa', 'EEE', 22, 'Delhi', 9345678901);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(7, 'Anil', 'Civil', 21, 'Chennai', 9234567890);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(8, 'Pooja', 'CSE', 23, 'Mumbai', 9123456789);
Query OK, 1 row affected (0.001 sec)

mysql> insert into Student_info values(9, 'Rahul', 'Mech', 20, 'Pune', 9345678123);
Query OK, 1 row affected (0.001 sec)
```

```
mysql> select * from Student_info;
```

StudentID	Name	Dept	Age	City	ContactNumber
1	Arjun	CSE	20	Hyderabad	38947589
2	Meena	ECE	21	Delhi	9123456780
2	Meena	ECE	21	Delhi	9123456780
3	Ravi	NULL	19	Chennai	9988776655
4	Sita	IT	NULL	Mumbai	9876501234
5	Kiran	CSE	20	Hyderabad	9876543211
6	Deepa	EEE	22	Delhi	9345678901
7	Anil	Civil	21	Chennai	9234567890
8	Pooja	CSE	23	Mumbai	9123456789
9	Rahul	Mech	20	Pune	9345678123

```
10 rows in set (0.001 sec)
```


Q10) Insert at least 2 records at once into the Student_info table, but only for the columns StudentID, Name, and Dept.

```
mysql> insert into Student_info(StudentID, Name, Dept) values
-> (10, 'Manoj', 'CSE'),
-> (11, 'Lakshmi', 'IT');
[Query OK, 2 rows affected (0.001 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

Q11) Write down the difference you observed between Q9 and Q10 in terms of:

- Which columns require values?

```
mysql> select * from Student_info;
+-----+-----+-----+-----+-----+-----+
| StudentID | Name   | Dept  | Age  | City   | ContactNumber |
+-----+-----+-----+-----+-----+-----+
| 1 | Arjun  | CSE   | 20   | Hyderabad | 38947589 |
| 2 | Meena  | ECE   | 21   | Delhi    | 9123456780 |
| 2 | Meena  | ECE   | 21   | Delhi    | 9123456780 |
| 3 | Ravi   | NULL  | 19   | Chennai  | 9988776655 |
| 4 | Sita   | IT    | NULL | Mumbai   | 9876501234 |
| 5 | Kiran  | CSE   | 20   | Hyderabad | 9876543211 |
| 6 | Deepa  | EEE   | 22   | Delhi    | 9345678901 |
| 7 | Anil   | Civil | 21   | Chennai  | 9234567890 |
| 8 | Pooja  | CSE   | 23   | Mumbai   | 9123456789 |
| 9 | Rahul  | Mech  | 20   | Pune     | 9345678123 |
| 10 | Manoj  | CSE   | NULL | NULL     | NULL      |
| 11 | Lakshmi | IT    | NULL | NULL     | NULL      |
+-----+-----+-----+-----+-----+-----+
12 rows in set (0.000 sec)
```

The rows with no values are stored as NULL.